

Literature Status for arXiv:math/0002219: Asymptotic ℓ_p FDDs

Run fa_banach_001, agent_lane_03

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Status

This is a `literature_already_answered` packet. It records that Problem 5.4 in Odell–Schlumprecht’s *Trees and Branches in Banach Spaces*, arXiv:math/0002219, is explicitly solved by Odell–Schlumprecht–Zsak, arXiv:math/0603063.

Source Question

On PDF page 24 of arXiv:math/0002219, in the final part of Section 5, the authors ask Problem 5.4. They define a reflexive space X to be asymptotic ℓ_p if there exists $K < \infty$ such that, for every k , every k -term member of the asymptotic structure $\{X\}_k$ is K -equivalent to the unit vector basis of ℓ_p^k . They also define an FDD (E_n) for a space Z to be asymptotic ℓ_p if all normalized k -block sequences of the tail $(E_i)_{i=k}^\infty$ are uniformly equivalent to ℓ_p^k .

The problem asks whether, if X is reflexive asymptotic ℓ_p for some $1 < p < \infty$, then X embeds into a space Z with an asymptotic ℓ_p FDD.

Supporting Answer

The abstract of arXiv:math/0603063 states the positive answer in essentially the same language: if X is separable, reflexive, and asymptotic ℓ_p , then X embeds into a reflexive space Z with an asymptotic ℓ_p finite-dimensional decomposition. The first page says this is proved as Corollary 4.9.

The explicit cross-reference appears on PDF page 32: immediately before Corollary 4.9, the paper says that a special case of Corollary 4.8 solves Problem 5.4 raised in the earlier Odell–Schlumprecht paper. Corollary 4.9 then states that every reflexive asymptotic ℓ_p space embeds into a reflexive space with an asymptotic ℓ_p FDD, and is also a quotient of such a reflexive space.

Scope

This is a positive literature answer to the source problem as stated. The supporting result is stronger than the source question because it gives a reflexive ambient space and a quotient formulation, and it is stated for the customary endpoint range as well. This packet does not claim a new proof, a new quantitative constant, or a strengthening beyond arXiv:math/0603063.

Search Evidence

The lane-3 queue selected arXiv:math/0002219. Cheap run indexes were searched for the arXiv id, source title, “0603063”, “asymptotic l_p spaces”, “asymptotic ℓ_p FDD”, and “Problem 5.4”; no duplicate packet for this target was found. Targeted web search for the exact embedding phrase identified arXiv:math/0603063, and the local parsed TeX verifies the explicit statement that Corollary 4.9 solves Problem 5.4 from the source paper.

References

- [1] E. Odell and Th. Schlumprecht, *Trees and Branches in Banach Spaces*, arXiv:math/0002219.
- [2] E. Odell, Th. Schlumprecht, and A. Zsak, *On the structure of asymptotic l_p spaces*, arXiv:math/0603063.